

Pitch Detection: Autocorrelation (Time Domain) vs Cepstrum (Frequency Domain)

ELL7420 Advanced Digital Signal Processing — Design Assignment 2

Naivedya Sahu · 2026EEY7565 · with project partner · October 2026

Result in one paragraph Two pitch detectors were implemented in MATLAB and run on the same frames: **autocorrelation** (time domain) and **cepstrum** (frequency domain). They share the 40 ms frames, the quiet-frame test, the peak picker and the median filter, and differ only in the curve that is searched for a peak. On the two recordings with a reference (600 frames) the cepstrum makes about half as many frame errors on clean speech, **3.5 % against 7.3 %** (21 against 44 frames). It misses fewer voiced frames, and its only two gross pitch errors fall in a creaky passage where the reference is itself unsure; the autocorrelation loses the low-pitched ending of s2 and doubles the pitch for four frames. In white noise the order reverses between 30 dB and 20 dB SNR: from 20 dB down to 5 dB the autocorrelation has the lower error (11.9 % against 14.2 % at 10 dB). Pitch tracks for s3.wav and s4.wav are given in Section 6 and Appendix A; the two methods agree within 5 % on 97.6 % (s3) and 93.3 % (s4) of the frames both call voiced.

1. Problem and data

Pitch detection estimates the fundamental frequency F_0 of a quasi-periodic signal. For speech it has two parts: decide whether a short segment is **voiced** (vocal folds vibrating, periodic) or **unvoiced** (noise-like or silent), and, if voiced, measure the period.

The assignment asks for one time-domain and one frequency-domain method, a comparison of the two on the recordings that have a reference, and pitch estimates for the two recordings that do not.

Recording	Reference	Voiced frames (reference)	Reference pitch range	Note
s1.wav	p1.mat	156 of 300	79–364 Hz, median 211 Hz	high-pitched voice, creaky ending
s2.wav	p2.mat	175 of 300	78–222 Hz, median 125 Hz	low-pitched voice
s3.wav	none	to be found	to be found	speech runs to the end of the file
s4.wav	none	to be found	to be found	

All four recordings are 3 s long, sampled at $f_s = 8$ kHz, 16 bit. The references hold one value every 10 ms (300 values); zero means unvoiced or silence.

One property of the reference matters later: **every reference value is exactly $8000/k$ for an integer k** (lags 22 to 102). The reference is therefore the output of some detector that measures the period in whole samples. Agreement with it is not quite the same as being right, and it cannot be tighter than the spacing of those values (5 Hz near 200 Hz, 14 Hz near 333 Hz).

2. What the two methods share

The two detectors are built to differ in one thing only. Everything in this section is common to both.

Item	Choice
Frame step	10 ms (80 samples), 300 frames per recording, matching the reference
Frame	40 ms (320 samples) centred on the middle of each 10 ms step, mean removed, Hamming window
Search range	lags 20–114 samples (2.5–14.25 ms), that is 400 Hz down to 70 Hz
Quiet-frame test	a frame whose RMS amplitude is below 0.008 is unvoiced
Peak picker	the earliest local peak that reaches 82 % of the tallest local peak, refined by a parabola
Voicing test	the height of that peak must reach a threshold: 0.50 (autocorrelation), 0.12 (cepstrum)
Smoothing	5-frame median on the voicing decision and on the pitch

Peak picker. Both curves can peak at the period and again at its multiples. Taking the *earliest* peak that is nearly as tall as the tallest one favours the true period over twice the period. With y_{-1}, y_0, y_{+1} the curve at the chosen integer lag $\hat{k}$ and its neighbours, a parabola through the three points moves the peak by

$$\delta = \frac{1}{2} \frac{y_{-1} - y_{+1}}{y_{-1} - 2y_0 + y_{+1}}, \quad \hat{F}_0 = \frac{f_s}{\hat{k} + \delta},$$

limited to half a sample.

Smoothing. A frame stays voiced only if most of the five frames around it are voiced. Its pitch becomes the middle value of the voiced ones among those five (the lower of the two middle values when their number is even, so the output is always a value that was actually measured). The scores are reported with and without this filter.

Error measures. With the reference as the answer key, each frame is one of four cases.

Measure	Definition
TP, FP, FN, TN	reference voiced and detector voiced; reference unvoiced but detector voiced; reference voiced but detector unvoiced; both unvoiced
Precision	TP / (TP + FP): of the frames called voiced, how many were
Recall	TP / (TP + FN): of the voiced frames, how many were found
MAE	mean absolute pitch difference over the TP frames
Within 20 %	TP frames whose pitch is within 20 % of the reference, counted over all reference-voiced frames
Frame error	share of all frames that are wrong in either way: FP, FN, or a TP frame more than 20 % off (a gross error)

3. Method A — autocorrelation (time domain)

3.1 Principle

Autocorrelation asks how well a frame matches a delayed copy of itself. At lag k the overlapping samples of the frame $y(n)$ and of its copy are multiplied and added, and the sum is divided by the energies of the two overlapping parts:

$$r(k) = \frac{\sum_{n=0}^{L-1-k} y(n) y(n+k)}{\sqrt{\sum_{n=0}^{L-1-k} y(n)^2 \sum_{n=0}^{L-1-k} y(n+k)^2}}, \quad L = 320.$$

If the frame repeats every P samples, the copy lines up with the original at $k = P$ and $r(k)$ has a peak there. The pitch is $F_0 = f_s/P$.

Dividing by both energies does two jobs. It removes loudness, so a quiet vowel and a loud one give the same peak height. It also removes the effect of the shrinking overlap: a plain sum has fewer terms at long lags and would fall away for that reason alone.

One effect remains. The Hamming window shapes the two overlapping parts differently, so even a perfectly periodic frame does not reach 1, and it reaches less the longer the lag. For an ideal periodic tone this detector gives $r = 0.94$ at 300 Hz, 0.86 at 150 Hz and 0.69 at 85 Hz. A low voice therefore starts closer to the 0.50 threshold than a high one. This shows up in the results (Section 5.3).

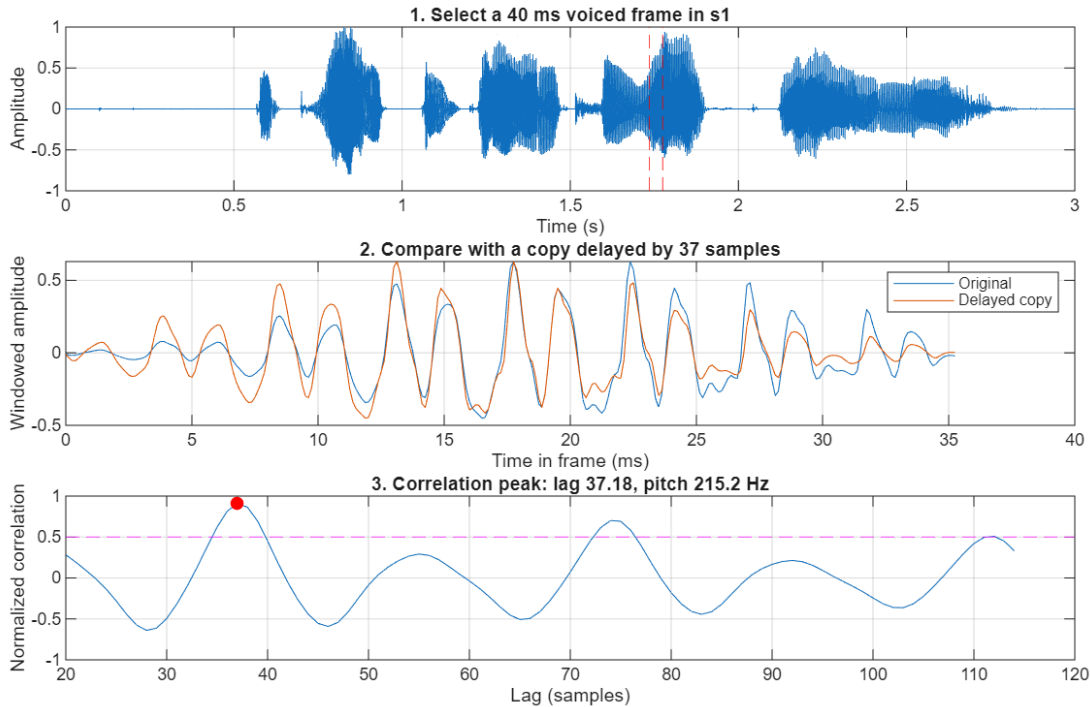


Figure 1. Autocorrelation on one voiced frame of s1 (near 1.755 s). Top: the 40 ms frame selected. Middle: the frame and its copy delayed by 37 samples; they nearly coincide. Bottom: normalised correlation against lag, with the 0.50 threshold (dashed); the peak at 37.18 samples gives 215.2 Hz.

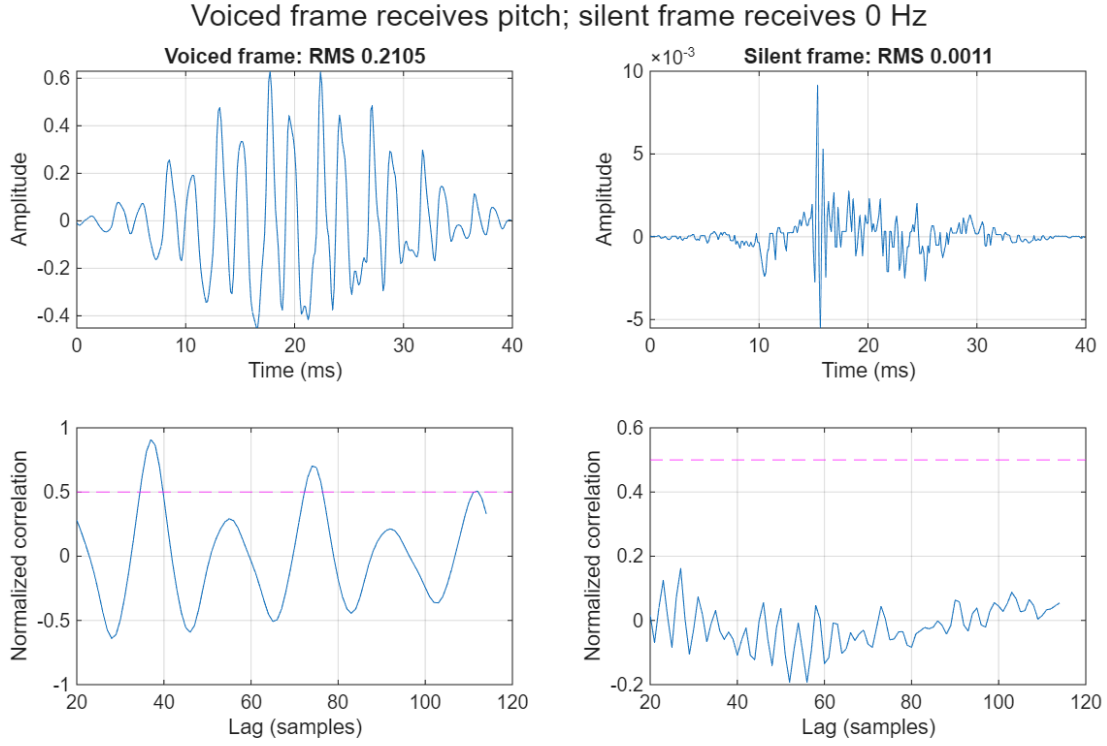


Figure 2. A voiced frame (left, RMS 0.2105) and a quiet frame (right, near 0.205 s, RMS 0.0011) of s1 with their correlation curves. The voiced frame has a peak well above 0.50. The quiet frame fails the RMS test and its curve has no such peak.

3.2 Steps and parameters

1. Take the 40 ms frame, remove its mean, apply the Hamming window.
2. Stop if the RMS is below 0.008: unvoiced.
3. Compute $r(k)$ for lags 20–114.
4. Pick the earliest local peak that reaches 82 % of the tallest, refine it with the parabola.
5. Voiced if that peak is at least 0.50; then $F_0 = f_s/\hat{k}$.

4. Method B — cepstrum (frequency domain)

4.1 Principle

Voiced speech is a train of pulses from the vocal folds, $e(n)$, shaped by the mouth and throat, $h(n)$. In the frequency domain the shaping is a multiplication, and the logarithm turns the multiplication into a sum:

$$x(n) = e(n) * h(n) \Rightarrow \log |X(f)| = \log |E(f)| + \log |H(f)|.$$

Because the pulses are regular, $|E(f)|$ is a comb of harmonics spaced F_0 apart, so $\log |E|$ is a **fast, regular ripple** along the frequency axis. The mouth's part, $\log |H|$, is a **slow** envelope with a few humps. An inverse Fourier transform of the log spectrum separates fast from slow. That transform is the real cepstrum [2]:

$$c(q) = \frac{1}{N} \sum_{m=0}^{N-1} \log |X(m)| e^{j2\pi mq/N}, \quad N = 1024.$$

Its variable q (quefrency) is a time, counted here in samples exactly like the lag of the autocorrelation. The slow envelope lands at small q , below the search range. A ripple that repeats every F_0 Hz lands as one sharp peak at $q = f_s/F_0$, the pitch period. An unvoiced frame has no harmonics, so no ripple and no peak.

One way to see why: $c(q)$ is the log spectrum multiplied by a cosine of the frequency axis, $\cos(2\pi f q)$, and added up. The crests of that cosine are $1/q$ Hz apart. When $1/q$ equals the harmonic spacing, every crest sits on a harmonic and everything adds; for any other q crests fall on harmonics and gaps alike and the sum cancels.

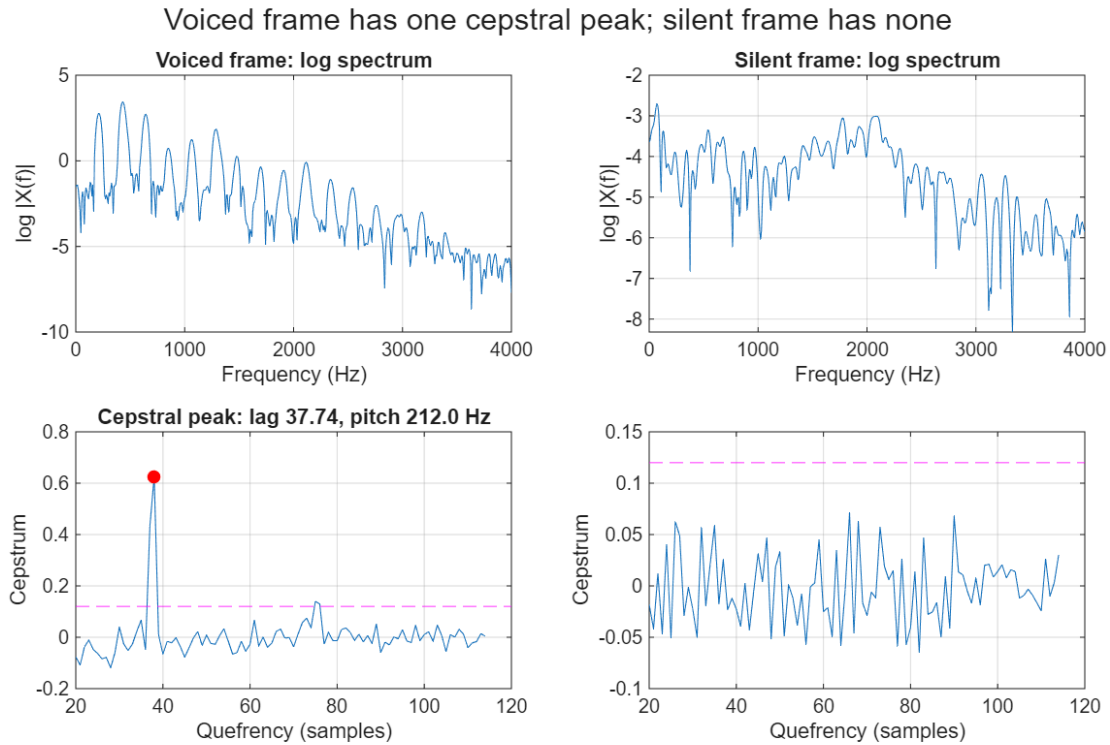


Figure 3. The two frames of Figure 2 through the cepstrum. Top: log magnitude spectra. The voiced frame (left) has a regular ripple, one bump per harmonic; the quiet frame (right) has none. Bottom: cepstra over the search range with the 0.12 threshold (dashed). The voiced frame has one sharp peak at 37.74 samples, 212.0 Hz, close to the 215.2 Hz found by the autocorrelation. The quiet frame stays below the threshold.

4.2 Steps and parameters

1. Take the same prepared frame as Method A (40 ms, mean removed, Hamming window).
2. Stop if the RMS is below 0.008: unvoiced.
3. 1024-point FFT, logarithm of the magnitude, inverse FFT. Keep quefrencies 20–114.
4. Pick the peak with the same rule as Method A.
5. Voiced if that peak is at least 0.12; then $F_0 = f_s/\hat{q}$.

4.3 One spectrum, two nonlinearities

The two methods are closer than their names suggest. By the Wiener–Khinchin relation the autocorrelation is the inverse Fourier transform of the power spectrum, so both curves are an inverse transform of a function of the magnitude spectrum:

$$R(k) = \mathcal{F}^{-1}\{|X(f)|^2\}, \quad c(q) = \mathcal{F}^{-1}\{\log |X(f)|\}.$$

- **Squaring** lets the strongest parts of the spectrum dominate. Those are the least affected by noise, but they are also the resonances of the mouth, which put peaks of their own into the curve.
- **The logarithm** levels the spectrum so that every harmonic counts about equally. The mouth's shape is pushed out of the way and the pitch peak is sharp and alone; but weak, noisy parts of the spectrum now count as much as strong ones.

With everything else held equal, as it is here, this one difference is what the results measure.

5. Results on the recordings with a reference (s1, s2)

5.1 Pitch tracks

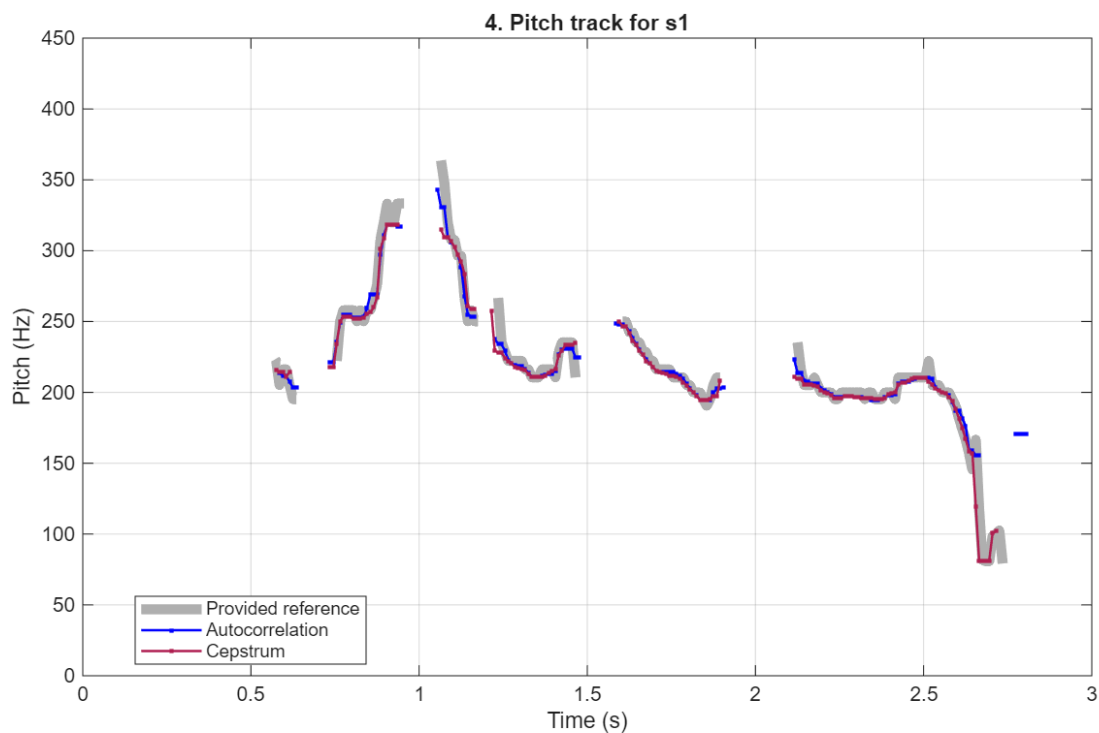


Figure 4. s1: both estimates (median-filtered) over the supplied reference.

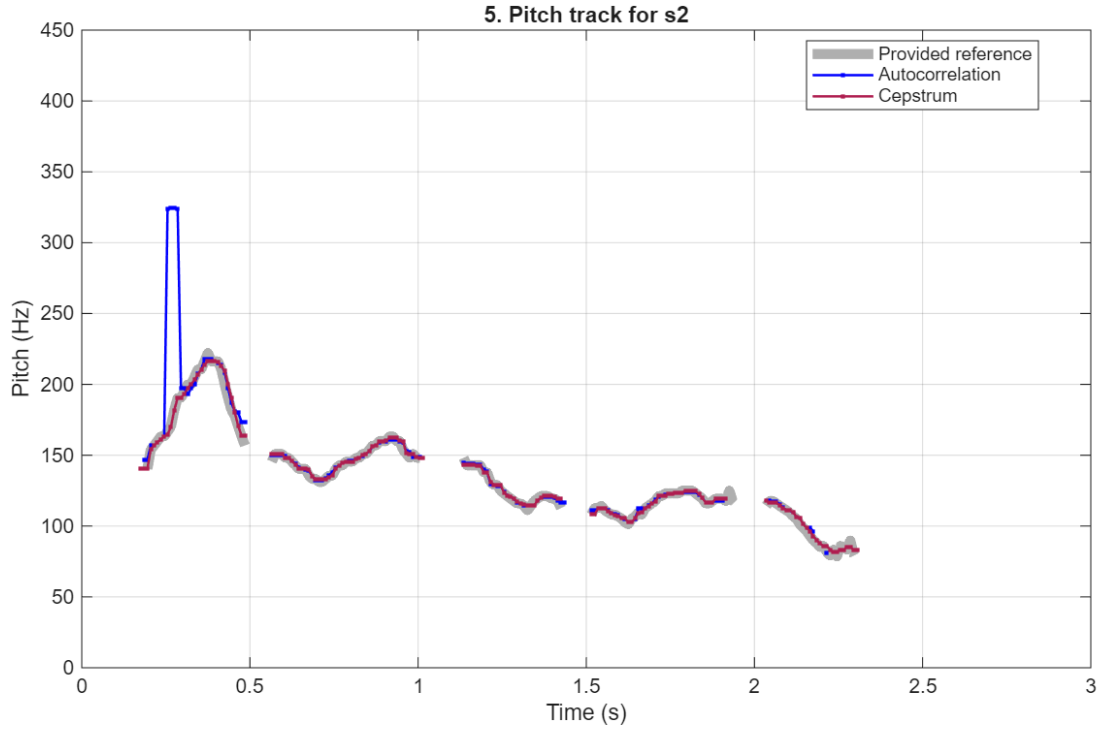


Figure 5. s2: both estimates over the supplied reference. The autocorrelation jumps to about 324 Hz near 0.26 s and stops early at the end.

Both tracks follow the reference through every voiced segment, including the fast rise and fall of s1 between 0.75 s and 1.2 s.

5.2 Scores

File	Method	Filter	TP	FP	FN	TN	Prec. %	Recall %	MAE (Hz)	Within 20 %	Error %
s1	Autocorrelation	none	147	13	9	131	91.9	94.2	7.16	143/156	8.7
s1	Autocorrelation	median	146	13	10	131	91.8	93.6	4.59	145/156	8.0
s1	Cepstrum	none	148	8	8	136	94.9	94.9	4.98	145/156	6.3
s1	Cepstrum	median	149	6	7	138	96.1	95.5	5.07	147/156	5.0
s2	Autocorrelation	none	160	2	15	123	98.8	91.4	5.69	156/175	7.0
s2	Autocorrelation	median	161	2	14	123	98.8	92.0	5.37	157/175	6.7
s2	Cepstrum	none	172	4	3	121	97.7	98.3	1.26	172/175	2.3
s2	Cepstrum	median	173	4	2	121	97.7	98.9	1.35	173/175	2.0
s1 + s2	Autocorrelation	median	307	15	24	254	95.3	92.7	5.00	302/331	7.3
s1 + s2	Cepstrum	median	322	10	9	259	97.0	97.3	3.07	320/331	3.5

"Filter" is the 5-frame median; "Error" is the frame error. Without the median filter the pooled frame errors are 7.8 % (autocorrelation) and 4.3 % (cepstrum).

5.3 Where the errors are

- **Missed voiced frames (FN).** 24 for the autocorrelation, 9 for the cepstrum. Eleven of the autocorrelation's are the low-pitched ending of s2 (frames 219–230, reference 78–90 Hz). There the correlation at the true lag is low, 0.17–0.42 in the four frames checked, below the 0.50 threshold. This is the window effect of Section 3.1: at such long lags even a perfect tone would only reach about 0.69. The cepstrum keeps all of these frames. Both methods lose part of the creaky ending of s1 (autocorrelation frames 268–274, cepstrum 273–274), where the reference itself jumps between 167, 118, 82 and 99 Hz from one frame to the next.
- **False voiced frames (FP).** 15 against 10. Most sit one or two frames outside a voiced segment, where the 40 ms window already reaches into the vowel. The autocorrelation also calls four frames after the end of s1 voiced (frames 278–281).
- **Gross pitch errors.** Five for the autocorrelation, two for the cepstrum. Four of the five are s2 frames 26–29 (0.255–0.285 s), reported near 324 Hz where the reference is 167–190 Hz: **double the pitch**. In those frames the correlation near half the true lag is 0.6–0.7, within 82 % of the true peak (0.69–0.78), so the "earliest strong peak" rule takes the half lag. The cause is a strong second harmonic. The cepstrum, given the same rule on the same frames, is not misled, because the logarithm has levelled that harmonic. Its two gross errors (s1 frames 266–267) and the autocorrelation's fifth (frame 267) are in the creaky ending.
- **Pitch value.** On matched frames the mean absolute error is 5.00 Hz against 3.07 Hz. The gap comes from the doubled frames: on s1, which has none, the two are level (4.59 and 5.07 Hz).
- **Median filter.** It helps both a little (7.8 % to 7.3 %, 4.3 % to 3.5 %). It removes isolated wrong values, three of the autocorrelation's eight raw gross errors among them, but a run of four wrong frames is too long for a 5-frame median.

6. Pitch estimates for s3 and s4

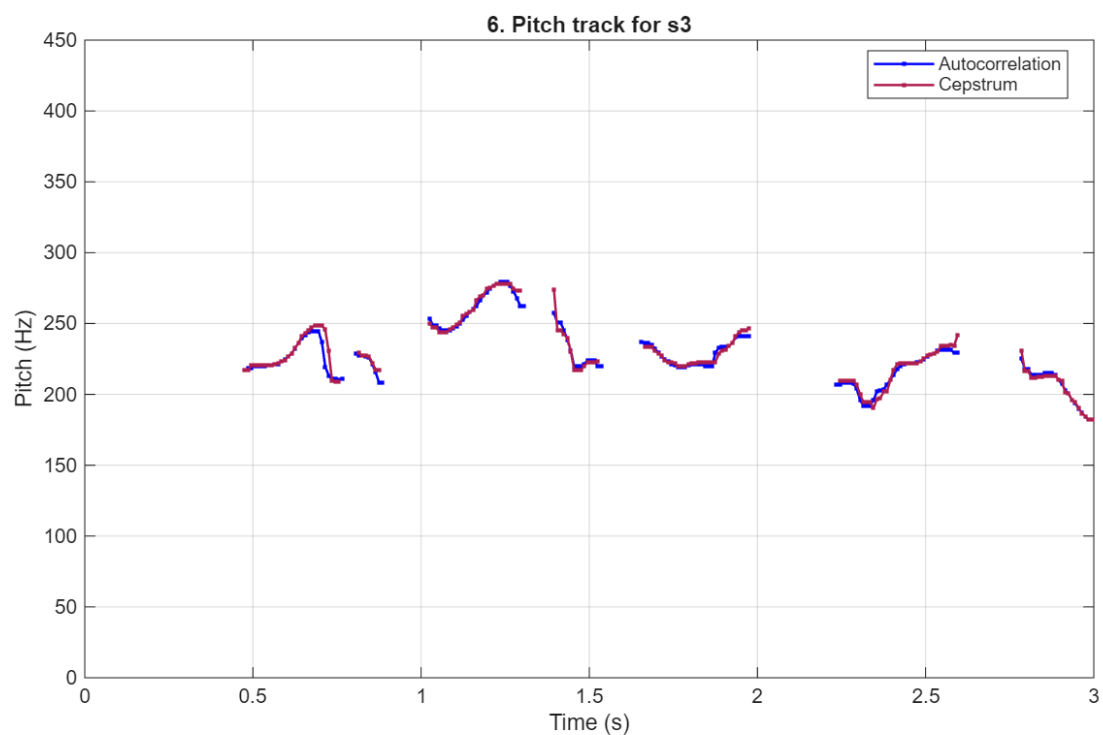


Figure 6. s3: both estimates. No reference exists.

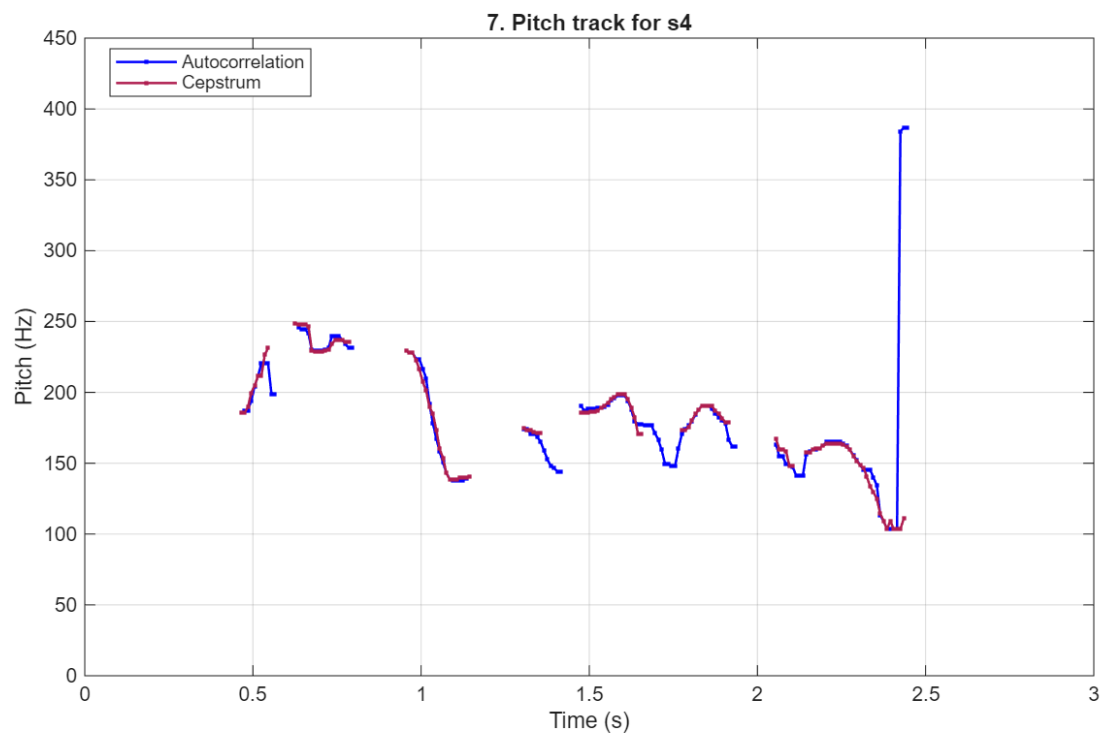


Figure 7. s4: both estimates. The autocorrelation track ends with three frames near 385 Hz.

File	Method	Voiced frames (of 300)	Median pitch
s3	Autocorrelation	174	223 Hz
s3	Cepstrum	168	223 Hz
s4	Autocorrelation	145	177 Hz
s4	Cepstrum	122	181 Hz

s3 is a high-pitched voice, about 180–280 Hz with a median of 223 Hz; its last voiced segment is cut off by the end of the file. **s4** is lower, about 105–250 Hz with a median near 180 Hz, ending in a fall to about 105 Hz.

The complete tracks are in `results/p3.mat` and `results/p4.mat` (variables `p3_autocorrelation`, `p3_cepstrum`, `p4_autocorrelation`, `p4_cepstrum`, each 300×1 in Hz with 0 for unvoiced, as in p1 and p2), in `results/s3_pitch.csv` and `results/s4_pitch.csv`, and in Appendix A.

With no reference, the two methods can only be checked against each other. Agreement is not accuracy, since both can be wrong together; but Section 5 showed that they fail in different places, and on s1 and s2 a frame on which they agree is almost always right.

File	Same voiced decision	Frames voiced in both	Within 5 % of each other	Mean difference
s1	94.0 %	148	95.9 %	2.9 Hz
s2	94.7 %	162	95.1 %	4.7 Hz
s3	97.3 %	167	97.6 %	2.2 Hz
s4	90.3 %	119	93.3 %	7.1 Hz

s3 is the easiest of the four recordings: the two tracks differ in the voiced decision on only 8 frames and never by more than 20 % in pitch. **s4 is the hardest**, and the disagreements are of two kinds.

- **The last three voiced frames of the autocorrelation (2.425–2.445 s)** read 384–387 Hz where the cepstrum reads 103–111 Hz (and unvoiced in the third). The pitch has been falling steadily to about 105 Hz, so the cepstrum is right; the autocorrelation has locked on to a lag at the short end of the search range.
- **26 frames that only the autocorrelation calls voiced**, mostly two stretches between syllables (1.365–1.415 s and 1.665–1.765 s), at 144–177 Hz in line with the pitch on either side. Judging by a narrow-band spectrogram, these are weak sounds with little energy above the lowest harmonics. The waveform still repeats, so the autocorrelation finds a period, but with so few harmonics there is no ripple for the cepstrum to find. Without a reference this cannot be settled, but the autocorrelation is probably right to call them voiced.

For s4 the safer track is therefore the cepstrum for the pitch values and the autocorrelation for these weak voiced stretches.

7. Noise

The recordings are clean. To see how each method behaves when they are not, white Gaussian noise was added to s1 and s2 at signal-to-noise ratios from 30 dB down to 0 dB (ratio of average powers over the whole recording), ten realisations each, with every threshold left as it was.

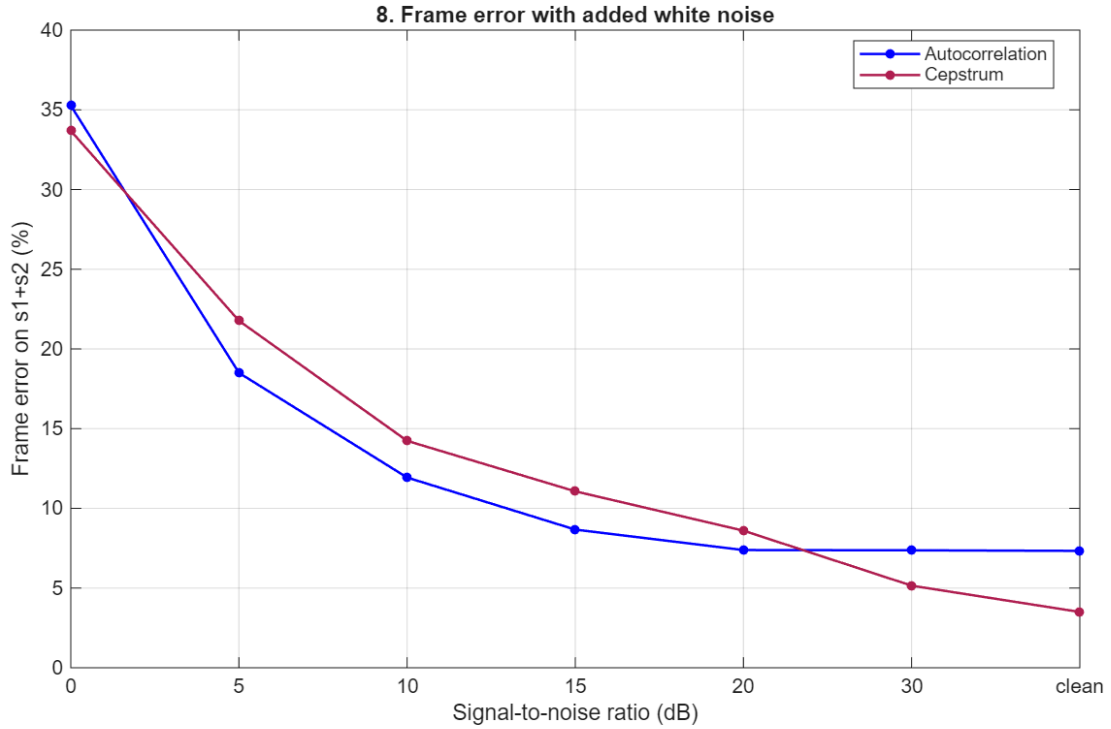


Figure 8. Frame error on s1 + s2 against SNR, median-filtered tracks, mean of ten trials.

SNR (dB)	clean	30	20	15	10	5	0
Autocorrelation frame error %	7.3	7.4	7.4	8.7	11.9	18.5	35.3
Cepstrum frame error %	3.5	5.1	8.6	11.1	14.2	21.8	33.7

- **The order reverses between 30 dB and 20 dB.** From 20 dB down to 5 dB the autocorrelation has the lower error. At 0 dB both have failed, with about a third of the frames wrong.
- **The autocorrelation barely notices noise down to 20 dB** (7.3 % to 7.4 %), while the cepstrum's error more than doubles (3.5 % to 8.6 %). This is the difference of Section 4.3. Squaring keeps the autocorrelation on the strong low harmonics, which stand well above the noise. The logarithm gives the gaps between harmonics, and the upper band where speech is weak, the same weight as the harmonics; once those hold noise the ripple is diluted and the cepstral peak sinks below a threshold that was set on clean speech.
- **Part of the loss is the quiet-frame test.** It compares against a fixed level. Below roughly 20 dB SNR the added noise alone exceeds that level, so silent frames are no longer removed by it and the peak-height threshold has to reject them unaided.

8. Summary of the comparison

Aspect	Autocorrelation	Cepstrum
Domain	time (lag)	frequency → quefrency
Curve searched	normalised correlation $r(k)$	$c(q)$, inverse FFT of the log spectrum
Frame error, clean speech (s1 + s2)	7.3 %	3.5 %
Precision / recall	95.3 % / 92.7 %	97.0 % / 97.3 %
Gross pitch errors	5 frames	2 frames
Frame error at 10 dB SNR	11.9 %	14.2 %
Typical failures	low pitch scores too low; doubles when the second harmonic is strong	weak sounds with few harmonics; noise
Cost per frame	95 short sums	two 1024-point FFTs and a logarithm per bin

9. Limitations

- **Small test set.** Two recordings, two speakers, 600 frames. One frame is 0.17 %. Errors cluster in a few places (the ending of s2, four frames of doubling, segment edges), so the gap of 23 frames is clear in direction but rests on a handful of events.
- **Thresholds chosen on the scored data.** The RMS level 0.008, the peak thresholds 0.50 and 0.12 and the 82 % rule were all set while looking at these recordings. The scores on s1 and s2 are therefore on the optimistic side, and other speakers would need the thresholds checked again.
- **The reference is not ground truth.** It is another detector's output, in whole-sample periods, with doubtful frames in the creaky ending of s1.
- **One window for both methods.** 40 ms suits the cepstrum, which needs several periods in the frame. An autocorrelation detector would usually use a shorter, less tapered window, and would then lose fewer low-pitched frames. The comparison is of the two methods under equal conditions, not of each at its best.
- **Noise model.** White Gaussian noise only, with thresholds fixed.
- **No tracking.** Each frame is decided on its own, with only a 5-frame median afterwards. The four doubled frames in s2 and the three wrong frames at the end of s4 would be removed by a tracker that penalises sudden octave jumps.

10. Conclusions

1. **Both methods work.** Each follows the reference through every voiced segment of s1 and s2, with precision and recall above 92 %.
2. **On clean speech the cepstrum is better: 3.5 % frame error against 7.3 %.** Given the same frames, the same quiet-frame test and the same peak picker, it misses fewer voiced frames (9 against 24) and makes fewer gross pitch errors (2 against 5).

3. **The reason is the logarithm.** It levels the spectrum, so a voiced frame gives one sharp peak at the period whatever the pitch, and the mouth's resonances do not compete with it. The autocorrelation's two characteristic errors here, doubling when the second harmonic is strong and under-scoring low voices, both come from working on the unlevelled waveform through a tapered window.
4. **In noise the autocorrelation is better.** From 20 dB to 5 dB SNR it has the lower frame error (11.9 % against 14.2 % at 10 dB), because the same logarithm gives noisy parts of the spectrum full weight.
5. **s3 and s4.** s3 has a pitch of about 180–280 Hz (median 223 Hz) and s4 about 105–250 Hz (median near 180 Hz). The two estimates agree within 5 % on 97.6 % and 93.3 % of the frames both call voiced. Where they part in s4, the cepstrum has the better pitch values and the autocorrelation probably the better voiced decision.
6. **Choice.** For clean recordings the cepstrum. For noisy ones the autocorrelation. Since they fail in different places, running both and flagging the frames on which they disagree is a cheap check, and it is what Section 6 does.

References

1. L. R. Rabiner, "On the use of autocorrelation analysis for pitch detection," *IEEE Trans. Acoustics, Speech, and Signal Processing*, vol. ASSP-25, no. 1, pp. 24–33, Feb. 1977.
2. A. M. Noll, "Cepstrum pitch determination," *J. Acoustical Society of America*, vol. 41, no. 2, pp. 293–309, 1967.
3. L. R. Rabiner, M. J. Cheng, A. E. Rosenberg and C. A. McGonegal, "A comparative performance study of several pitch detection algorithms," *IEEE Trans. Acoustics, Speech, and Signal Processing*, vol. ASSP-24, no. 5, pp. 399–418, Oct. 1976.
4. L. R. Rabiner, M. R. Sambur and C. E. Schmidt, "Applications of a nonlinear smoothing algorithm to speech processing," *IEEE Trans. Acoustics, Speech, and Signal Processing*, vol. ASSP-23, no. 6, pp. 552–557, Dec. 1975.

Appendix A. Pitch estimates for s3.wav and s4.wav

One value per 10 ms frame, in Hz, rounded to the nearest hertz, after the median filter. A dot is an unvoiced or silent frame. The row label is the start time of the row; columns step by 10 ms. Full precision is in `results/s3_pitch.csv` and `results/s4_pitch.csv`.

s3.wav, autocorrelation (Hz)

t (s)	+0	+10	+20	+30	+40	+50	+60	+70	+80	+90 ms
0.0	.	.	.	.	.	.	.	.	.	.
0.1	.	.	.	.	.	.	.	.	.	.
0.2	.	.	.	.	.	.	.	.	.	.
0.3	.	.	.	.	.	.	.	.	.	.
0.4	.	.	.	.	.	.	.	.	219	219
0.5	220	220	220	220	220	220	221	222	223	225
0.6	226	229	232	236	239	242	244	245	245	245
0.7	237	220	213	211	211	210	211	.	.	.
0.8	229	228	228	226	226	221	216	208	208	.
0.9	.	.	.	.	.	.	.	.	.	.
1.0	.	.	254	249	249	246	245	245	245	246
1.1	248	250	253	255	258	260	262	266	270	272
1.2	275	277	278	279	279	279	277	273	268	262
1.3	262	.	.	.	.	.	.	.	.	257
1.4	251	251	245	239	231	220	220	220	221	224
1.5	224	224	220	220	.	.	.	.	.	.
1.6	.	.	.	.	.	237	236	236	235	233
1.7	230	227	225	223	222	220	219	219	219	220
1.8	221	221	221	221	220	220	220	229	233	233
1.9	234	234	236	240	241	241	241	241	.	.
2.0	.	.	.	.	.	.	.	.	.	.
2.1	.	.	.	.	.	.	.	.	.	.
2.2	.	.	.	207	207	208	208	208	208	204
2.3	196	192	192	192	196	202	203	204	207	210
2.4	214	218	220	221	222	222	222	222	224	225
2.5	227	228	229	231	231	231	231	231	229	229
2.6	.	.	.	.	.	.	.	.	.	.
2.7	.	.	.	.	.	.	.	.	225	218
2.8	218	214	214	214	214	215	215	215	214	210
2.9	207	203	200	196	194	190	187	184	182	182

s3.wav, cepstrum (Hz)

t (s)	+0	+10	+20	+30	+40	+50	+60	+70	+80	+90 ms
0.0	.	.	.	.	.	.	.	.	.	.
0.1	.	.	.	.	.	.	.	.	.	.
0.2	.	.	.	.	.	.	.	.	.	.
0.3	.	.	.	.	.	.	.	.	.	.
0.4	.	.	.	.	.	.	.	217	217	220
0.5	220	220	220	220	220	221	221	222	223	224
0.6	227	229	233	236	241	243	245	248	249	249
0.7	249	246	231	210	209	209	.	.	.	.
0.8	.	229	227	227	227	222	217	217	.	.
0.9	.	.	.	.	.	.	.	.	.	.
1.0	.	.	250	247	247	244	244	244	246	247
1.1	249	250	255	257	258	259	267	269	270	274
1.2	275	277	278	278	278	278	278	275	273	273
1.3	.	.	.	.	.	.	.	.	.	274
1.4	245	245	242	240	230	217	217	217	220	223
1.5	223	223	223	.	.	.	.	.	.	.
1.6	.	.	.	.	.	.	234	234	234	231
1.7	229	227	224	223	223	222	220	220	220	221
1.8	222	222	222	222	222	222	223	223	229	231
1.9	232	234	236	241	244	245	245	247	.	.
2.0	.	.	.	.	.	.	.	.	.	.
2.1	.	.	.	.	.	.	.	.	.	.
2.2	.	.	.	.	209	209	210	210	210	207
2.3	200	194	194	194	190	197	197	202	202	210
2.4	217	221	222	222	222	222	222	222	223	225
2.5	227	228	229	230	234	234	234	235	234	242
2.6	.	.	.	.	.	.	.	.	.	.
2.7	.	.	.	.	.	.	.	.	230	216
2.8	216	212	212	212	212	213	213	213	213	210
2.9	209	201	201	196	194	190	187	185	182	182

s4.wav, autocorrelation (Hz)

t (s)	+0	+10	+20	+30	+40	+50	+60	+70	+80	+90 ms
0.0	.	.	.	.	.	.	.	.	.	.
0.1	.	.	.	.	.	.	.	.	.	.
0.2	.	.	.	.	.	.	.	.	.	.
0.3	.	.	.	.	.	.	.	.	.	.
0.4	.	.	.	.	.	.	.	187	187	194
0.5	204	211	221	221	221	199	199	.	.	.
0.6	.	.	.	246	245	245	242	230	230	229
0.7	230	230	232	240	240	240	237	234	231	231
0.8	.	.	.	.	.	.	.	.	.	.
0.9	.	.	.	.	.	230	228	228	223	223
1.0	217	210	192	178	167	158	151	143	138	138
1.1	138	138	138	139	.	.	.	.	.	.
1.2	.	.	.	.	.	.	.	.	.	.
1.3	174	174	171	171	169	165	159	153	148	146
1.4	144	144	.	.	.	.	.	191	187	188
1.5	188	188	189	189	190	191	194	196	198	198
1.6	198	194	188	179	177	177	177	177	177	171
1.7	167	159	150	150	148	148	160	170	174	177
1.8	180	184	188	191	191	191	188	185	183	180
1.9	178	166	162	162	.	.	.	.	.	.
2.0	.	.	.	.	.	163	155	155	149	148
2.1	147	141	141	141	156	159	159	160	161	163
2.2	165	165	165	165	165	164	162	160	156	152
2.3	149	145	145	145	140	134	113	109	104	104
2.4	104	104	384	387	387	.	.	.	.	.
2.5	.	.	.	.	.	.	.	.	.	.
2.6	.	.	.	.	.	.	.	.	.	.
2.7	.	.	.	.	.	.	.	.	.	.
2.8	.	.	.	.	.	.	.	.	.	.
2.9	.	.	.	.	.	.	.	.	.	.

s4.wav, cepstrum (Hz)

t (s)	+0	+10	+20	+30	+40	+50	+60	+70	+80	+90 ms
0.0	.	.	.	.	.	.	.	.	.	.
0.1	.	.	.	.	.	.	.	.	.	.
0.2	.	.	.	.	.	.	.	.	.	.
0.3	.	.	.	.	.	.	.	.	.	.
0.4	.	.	.	.	.	.	185	185	190	200
0.5	205	211	211	226	231	.	.	.	.	.
0.6	.	.	248	248	248	248	247	229	229	229
0.7	229	229	230	234	237	237	237	235	235	.
0.8	.	.	.	.	.	.	.	.	.	.
0.9	.	.	.	.	.	229	228	228	223	216
1.0	207	201	190	185	174	160	154	143	138	138
1.1	138	140	140	140	140	.	.	.	.	.
1.2	.	.	.	.	.	.	.	.	.	.
1.3	174	174	174	172	171	171	.	.	.	.
1.4	.	.	.	.	.	.	.	186	186	186
1.5	186	186	187	189	191	193	195	197	199	199
1.6	199	195	189	182	170	170	.	.	.	.
1.7	.	.	.	.	.	.	.	173	173	175
1.8	180	185	188	191	191	191	190	187	185	182
1.9	179	179	.	.	.	.	.	.	.	.
2.0	.	.	.	.	.	167	160	160	159	148
2.1	148	.	.	.	158	158	160	160	161	162
2.2	163	164	164	164	164	163	161	159	155	152
2.3	148	147	141	133	130	125	114	109	103	109
2.4	103	103	103	111	.	.	.	.	.	.
2.5	.	.	.	.	.	.	.	.	.	.
2.6	.	.	.	.	.	.	.	.	.	.
2.7	.	.	.	.	.	.	.	.	.	.
2.8	.	.	.	.	.	.	.	.	.	.
2.9	.	.	.	.	.	.	.	.	.	.

Appendix B. Files, reproduction and checks

File	Content
D2_code_2026EEY7565.mlx	MATLAB Live Script: both detectors, scoring, noise test, all figures, with explanations and outputs
results/s1_pitch.csv ... s4_pitch.csv	time, both estimates and the reference, frame by frame
results/p3.mat, results/p4.mat	pitch tracks for s3 and s4, both methods
results/report_plots/01 ... 08	the figures of this report

MATLAB R2025b, no toolbox needed. Open the Live Script and press Run, with `speech_samples/` beside it or one level up. The random seed is fixed, so the noise test reproduces exactly.

Two checks were made on the code:

- **The autocorrelation detector is unchanged by the integration.** Its functions were copied line for line, and its unsmoothed scores in Section 5.2 equal those of the stand-alone version (s1: TP 147, FP 13, FN 9, TN 131; s2: TP 160, FP 2, FN 15, TN 123).
- **The results were reproduced by an independent implementation.** The whole chain (framing, both curves, peak picker, thresholds, median filter) was written a second time in Python (NumPy). All eight pitch tracks agree with the Live Script: no frame differs in the voiced decision, and the pitch values agree to 10^{-12} Hz.